

VocalCraft: AI-Driven Pamphlet Creation Through Voice Commands

Project Team

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Session 2021-2025

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13th September, 2024

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Chapter 1

Introduction

VocalCraft is an AI-driven pamphlet creation application that helps users create promotional and menu pamphlets. This paper is intended to assist developers, testers, and supervisors in clarifying aspects of the architectural and modular structure of the project.

The concept of VocalCraft was inspired by the problems the existing image generation models had for generating the correct text on the images. The present-day image generation models superimpose the text on the image and then work on it as an image rather than a text, hence creating misleading images.

This paper discusses the multi-model architecture used by VocalCraft and the application system design, unlike currently available tools that solely rely on diffusing models resulting in low-quality pamphlet generation. This paper is intended for the developers and tester who understand the technical terms and diagrams helping them in understanding the design models of the system.

1.1 Problem Statement

Fast-food companies find it tough to produce promotional materials that might strike a chord. The current popular image generation models deliver visually good-looking content but often get the text wrong; this makes it unsuitable for marketing. Quite right, this is a problem for businesses, particularly those who cannot fix such errors to produce commercial quality material.

The proposed project will attempt to address this issue by designing a system that can automatically produce aesthetically appealing, integrated pamphlets with proper, professional text. This system shall not be complicated but rather simple to use in such a way that it will serve business people with a reliable mechanism of composing high-class promotional material for competition and customer interaction.

1.2 Scope

The scope of the **VocalCraft** project includes:

- **Voice Command Functionality:** This comprises a voice recognition model which is to transcribe the user's voice instructions in controlling the aspects of the pamphlet design such as layout, images, and text content.
- **Image Generation using Diffusion Model:** It integrates the Diffusion Model, which automates the generation of visual and textual content to make pamphlets attractive-looking with correctly and professionally styled text according to user prompts.
- **Customizable Templates:** To this end, the application will grant users the ability to edit and finalize pamphlets that have been generated according to their brand identity and marketing requirements.
- **Print-Ready Output:** With the application, the user can create pamphlets ready for printing. These could be exported into numerous formats such that sharing is easy, even in SVG format.
- **Cloud-Based Storage:** This lets the user save their projects online. Cloud solutions enable them to access and manipulate their projects from anywhere; thus, much convenience is achieved with such an option.
- **Mobile App Accessibility:** It will make the system accessible through the use of mobile applications. The system will therefore have a wider usability scale for different users.
- **User-Friendly Interface:** Creating a user-first interface that will ease the process of creating pamphlets, making it friendly for users with minimum design experience to produce quality promotional materials.

1.3 Modules

The following comprises of modules of VocalCraft

1.3.1 Voice Input and Transcription Module

Purpose: Convert the user's voice input into text for further processing.

Key Functionality:

- Capture voice input from the user through a mobile app.
- Transcribe speech to text.
- Ensure low-latency and high-accuracy transcription.

1.3.2 Prompt Engineering and NER Module

Purpose: Perform prompt engineering and extract key entities for pamphlet design.

Key Functionality:

- Extract important details (e.g., colors, text, layout) using Named Entity Recognition (NER).
- Store extracted entities to guide image generation and layout.

1.3.3 Image Generation and Inpainting Module

Purpose: Generate the pamphlet image with placeholders for text and images using inpainting.

Key Functionality:

- Use the diffusion model to generate the main pamphlet image based on NER data.
- Incorporate inpainting to add placeholders for text and object areas.
- Inference-time inpainting using a binary mask to place text and objects precisely.

1.3.4 Segmentation and Binary Mask Generation Module

Purpose: Create a binary mask to define areas for text and images on the pamphlet.

Key Functionality:

- Generate a binary mask for the text and image areas.
- Remove the objects from the final output image using
- Feed this mask into the diffusion model for inpainting.

1.3.5 Image Resizing and Optimization Module

Purpose: Resize and optimize the final image for printing or digital display.

Key Functionality:

- Resize the generated pamphlet based on user-specified print dimensions.
- Ensure image quality and aspect ratio are maintained during resizing.
- Provide options for different printing formats.

1.3.6 Mobile App Front-End Module

Purpose: Provide a user-friendly interface for voice input, pamphlet generation, and editing options.

Key Functionality:

- Capture voice input and manage user interaction.
- Display progress and final generated pamphlet.
- Allow users to edit generated pamphlets.
- Allow users to print pamphlets.

1.3.7 Backend and Cloud Integration Module

Purpose: Handle voice processing, model inference, and image generation in the cloud.

Key Functionality:

- Process voice input and prompts on the server.
- Run inference models in the cloud.
- Manage data flow between the app, models, and storage.

1.3.8 Deployment Phase

Purpose: Deploy the app for real-world use.

Key Functionality:

- Deploy the app on a server.
- Set up continuous deployment for future updates.
- Monitor app performance and user feedback for improvements.

1.4 User Classes and Characteristics

The following are the user classes that will use this product.

User Class	Description
End User (Regular User)	The End User is the intended specific user of the system, which often is a person wanting to create posters for individual use, advertising, or aesthetic. They will work with the system to as an example offer voice or text commands to capture prompts and review the created posters for modification. The users can also generate posters they have saved, shared or even those they want to retrieve from the list. End users come in all categories from those who do not have any clue on technical aspects to designers on the look for creative assistance.
Admin	The Admin manages the entire system. These are related to such processes as system performance control, account administration, error handling, machine learning model updates (such as Whisper or Stable Diffusion), and system security.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of VocalCraft AI.

2.1 Use-case

The following are the use cases of VocalCraft AI. The actors for this project are the **Users (Primary Actor)** who will be generating the posters, the **Admin (Secondary Actor)** who will manage the backend of the application, and the **System** that will process and generate the final output.

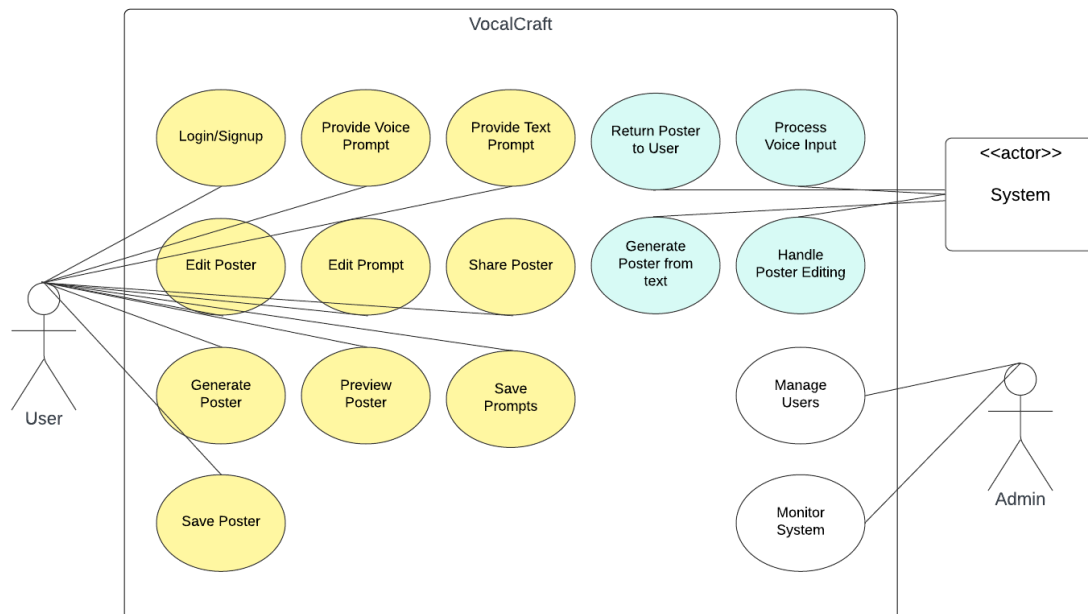


Figure 2.1: Use Case Diagram

2.1.1 User Authentication Sign Up/Log In

Field	Details
Use Case Name	User Authentication (Sign Up / Log In)
Scope	A user signs up or logs into the app.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to access app features after authentication.
Preconditions	The user is not authenticated.
Post-conditions	The user is authenticated and redirected to the dashboard.
Main Success Scenario	1. The user enters their credentials (or signs up with new details). 2. The system verifies the credentials or creates a new account. 3. The user is successfully logged in and redirected to the dashboard.

Table 2.1: User Authentication Use Case

2.1.2 Provide Voice Prompt

Field	Details
Use Case Name	Provide Voice Prompt
Scope	A user provides voice input to describe the poster they want to generate.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to describe the poster using voice input for efficient interaction. System: Captures and converts voice input to text accurately for further processing.
Preconditions	User is authenticated and given permission to access the microphone.
Post-conditions	Voice prompt is converted to text using the ASR (Automatic Speech Recognition) model.
Main Success Scenario	1. The user opens the app and navigates to the voice prompt input. 2. The user provides the voice input for the poster. 3. The system captures the voice input and processes it using the Whisper model for transcription. 4. The transcribed text is now successfully created and can proceed with the poster creation.

Table 2.2: Provide Voice Prompt Use Case

2.1.3 Provide Text Prompt

Field	Details
Use Case Name	Provide Text Prompt
Scope	The user provides a text prompt to describe the poster they want to generate.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to generate a poster using a custom text prompt. System: Accurately processes the text prompt and generates the corresponding poster.
Preconditions	User is authenticated and the app is ready to receive input.
Post-conditions	The text prompt is processed, and the system generates the corresponding poster.
Main Success Scenario	1. The user opens the app and navigates to the text input field. 2. The user types in a description for the poster. 3. The system processes the text prompt and sends it to the image generation model (Stable Diffusion). 4. The system successfully generates and displays the poster based on the text prompt.

Table 2.3: Provide Text Prompt Use Case

2.1.4 Generate Poster

Field	Details
Use Case Name	Generate Poster
Scope	The system generates a poster based on a voice or text prompt provided by the user.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to generate a visually appealing poster based on their input. System: Needs to accurately process the input prompt and generate a poster using the Stable Diffusion model.
Preconditions	The user has provided a valid voice or text prompt. The system uses the API for the poster generation model (Stable Diffusion).
Post-conditions	The poster is successfully generated and displayed for the user. The user can then proceed to edit or save the poster.
Main Success Scenario	1. The user provides a voice or text prompt describing the poster. 2. The system sends the prompt to the backend, which processes it through the Stable Diffusion model. 3. The system retrieves the generated poster and displays it to the user. 4. The user can view the poster and decide to save, edit, or regenerate it.

Table 2.4: Generate Poster Use Case

2.1.5 Preview Poster

Field	Details
Use Case Name	Preview Poster
Scope	The user previews the generated poster before finalizing it for further actions like editing, saving, or sharing.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to ensure the poster meets expectations before editing, saving, or sharing. System: Provides a visual representation of the poster for the user to review.
Preconditions	The user has already provided a voice or text prompt, and the system has generated a poster. The generated poster is ready for display.
Post-conditions	The user successfully views the poster and decides to either edit, save, share, or regenerate it.
Main Success Scenario	1. The system generates the poster based on the user's prompt. 2. The user is presented with a preview of the generated poster. 3. The user reviews the poster and decides whether to proceed with editing, saving, or sharing. 4. The system allows the user to take the next action (edit, save, share, or regenerate).

Table 2.5: Preview Poster Use Case

2.1.6 Edit Poster

Field	Details
Use Case Name	Edit Poster
Scope	The user can edit the generated poster by adjusting text, colors, fonts, or repositioning elements.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to customize the poster to better fit their preferences. System: Needs to provide a flexible editing interface to allow the user to modify the poster easily.
Preconditions	The user has already generated or previewed the poster. The system has loaded the poster and editing tools are available.
Post-conditions	The poster is modified and saved with the user's changes, or the user discards the changes. The updated poster is ready for saving or sharing.
Main Success Scenario	1. The user selects the edit option after generating or previewing the poster. 2. The system loads the editing interface, allowing the user to change text, colors, fonts, or reposition elements. 3. The user makes the desired modifications to the poster. 4. The user saves the changes, and the system updates the poster with the modifications. 5. The poster is ready for further actions such as saving or sharing.

Table 2.6: Edit Poster Use Case

2.1.7 Edit Prompt

Field	Details
Use Case Name	Edit Prompt
Scope	The user can modify the text or voice prompt previously provided for poster generation.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to refine or change the input prompt to generate a different poster. System: Needs to accept and process the edited prompt to generate a new poster.
Preconditions	The user has already provided an initial prompt (either voice or text). The system has generated a poster based on the initial prompt.
Post-conditions	The system updates the poster based on the edited prompt. The new poster is available for preview, editing, or saving.
Main Success Scenario	1. The user selects the option to edit the previously provided prompt. 2. The system presents the existing prompt (text or voice transcription) for modification. 3. The user modifies the prompt and submits it. 4. The system processes the edited prompt and generates a new poster. 5. The system displays the new poster for user preview or further action.

Table 2.7: Edit Prompt Use Case

2.1.8 Share Poster

Field	Details
Use Case Name	Share Poster
Scope	The user shares the generated poster via social media, email, or messaging apps.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to share the generated poster with others. System: Needs to provide options for sharing the poster via different platforms.
Preconditions	The user has generated or edited the poster and is satisfied with the final result. The poster is available for sharing, and the user is authenticated to use sharing features.
Post-conditions	The poster is successfully shared via the chosen platform (e.g., social media, email, or messaging app).
Main Success Scenario	1. The user selects the share option for the generated poster. 2. The system presents sharing options (e.g., social media, email, or messaging apps). 3. The user selects a platform for sharing. 4. The system shares the poster via the chosen platform, and the user receives confirmation of successful sharing.

Table 2.8: Share Poster Use Case

2.1.9 Save Poster

Field	Details
Use Case Name	Save Poster
Scope	The user saves the generated poster to their local device or cloud storage.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to save the generated poster for future use or further modifications. System: Needs to provide the ability to save the poster in a format usable on different platforms (e.g., PNG, JPEG).
Preconditions	The user has generated or edited the poster and is satisfied with the result. The poster is available for saving, and the user has storage permission on their device.
Post-conditions	The poster is successfully saved to the user's device or cloud storage in the selected format (e.g., PNG, JPEG).
Main Success Scenario	1. The user selects the save option for the generated poster. 2. The system provides format options for saving (e.g., PNG, JPEG). 3. The user selects a location on their device or cloud storage to save the poster. 4. The system confirms that the poster has been successfully saved in the selected location and format.

Table 2.9: Save Poster Use Case

2.1.10 Save Prompts

Field	Details
Use Case Name	Save Prompts
Scope	The user saves a voice or text prompt for future use in generating posters.
Level	User Goal
Primary Actor	User
Stakeholders and interests	User: Wants to save a prompt to reuse it for generating future posters without having to input it again. System: Needs to store the prompt securely and provide options to retrieve and reuse it.
Preconditions	The user has successfully provided a valid prompt (text or voice). The system is capable of saving user inputs and associating them with the user's account.
Post-conditions	- The prompt is saved in the system and associated with the user's profile. - The user can retrieve and reuse the saved prompt for future poster generation.
Main Success Scenario	1. The user selects the option to save a prompt after providing it. 2. The system presents a confirmation of the prompt to be saved. 3. The user confirms the save action. 4. The system saves the prompt in the user's account for future use. 5. The user can view or retrieve the saved prompt when needed.

Table 2.10: Save Prompts Use Case

2.1.11 Return Poster To User

Field	Details
Use Case Name	Return Poster to User
Scope	The system returns the generated poster to the user for pre-view, editing, or saving.
Level	System Goal
Primary Actor	System
Stakeholders and interests	User: Wants to receive the generated poster for further actions like editing, saving, or sharing. System: Needs to deliver the poster promptly and accurately based on the prompt provided by the user.
Preconditions	The user has provided a valid prompt (text or voice) and the system has successfully generated a poster.
Post-conditions	The generated poster is displayed to the user for preview and further actions.
Main Success Scenario	1. The user provides a voice or text prompt to the system. 2. The system processes the input and generates a poster based on the prompt. 3. The system sends the poster back to the user interface. 4. The user views the poster and proceeds to either edit, save, or share it.

Table 2.11: Return Poster to User Use Case

2.1.12 Process Voice Input

Field	Details
Use Case Name	Process Voice Input
Scope	The system processes the user's voice input and converts it into text using an ASR (Automatic Speech Recognition) model.
Level	System Goal
Primary Actor	System
Stakeholders and interests	User: Wants their voice input converted to text accurately for generating a poster. System: Needs to process the voice input efficiently and return accurate text for further poster generation steps.
Preconditions	The user has provided a valid voice input through the system's interface. The system has access to the ASR model (Whisper) for transcription.
Post-conditions	The voice input is successfully converted into text and made available for poster generation.
Main Success Scenario	1. The user speaks into the app to describe the poster. 2. The system captures the voice input and sends it to the ASR model. 3. The ASR model processes the voice input and transcribes it into text. 4. The system retrieves the transcribed text and displays it for the user to confirm or edit. 5. The text is ready for the next steps in poster generation.

Table 2.12: Process Voice Input Use Case

2.1.13 Generate Poster From Text

Field	Details
Use Case Name	Generate Poster From Text
Scope	The system generates a poster based on the text input provided by the user.
Level	System Goal
Primary Actor	System
Stakeholders and interests	User: Wants to see a poster generated based on the text prompt they provided. System: Needs to process the text input and generate an accurate poster using a machine learning model like Stable Diffusion.
Preconditions	The user has provided a valid text prompt describing the poster they want to generate. The system has access to the image generation model (Stable Diffusion).
Post-conditions	The poster is generated and displayed for the user to preview or edit.
Main Success Scenario	1. The user submits a text prompt describing the desired poster. 2. The system processes the text and sends it to the image generation model. 3. The model generates a poster based on the provided text. 4. The system retrieves the generated poster and displays it to the user. 5. The user can then preview or edit the poster before finalizing it.

Table 2.13: Generate Poster From Text Use Case

2.1.14 Handle Poster Editing

Field	Details
Use Case Name	Handle Poster Editing
Scope	The system processes user input to modify the generated poster, such as adjusting text, colors, fonts, or repositioning elements.
Level	System Goal
Primary Actor	System
Stakeholders and interests	User: Wants to edit the poster by making adjustments to its layout, text, and other visual elements. System: Must handle the user's input efficiently and update the poster in real-time, reflecting all changes accurately.
Preconditions	The user has generated a poster and initiated the editing process. The system provides an editing interface with tools for modifying the poster.
Post-conditions	The poster is updated based on the user's modifications and saved for further actions, such as saving or sharing.
Main Success Scenario	1. The system presents the user with tools for editing the poster. 2. The user interacts with the system to modify text, fonts, colors, or reposition elements. 3. The system processes the user's input in real time and updates the poster accordingly. 4. The system saves the modified poster for further actions, such as previewing, saving, or sharing.

Table 2.14: Handle Poster Editing Use Case

2.1.15 Manage Users

Field	Details
Use Case Name	Manage Users
Scope	The admin manages user accounts, including adding, removing, or modifying user accounts.
Level	User Goal
Primary Actor	Admin
Stakeholders and interests	Admin: Wants to manage user access, ensure only authorized users can access the system, and maintain system security. System: Needs to keep the database of users updated and ensure that appropriate permissions are in place for user actions.
Preconditions	The admin is authenticated and has the necessary permissions
Post-conditions	The user database is updated with the changes (add, remove, modify users) made by the admin.
Main Success Scenario	1. The admin logs into the system and navigates to the user management section. 2. The admin views the list of current users. 3. The admin adds a new user by filling in the necessary information. 4. The admin removes a user or modifies an existing user's 5. The system updates the user database accordingly and confirms the changes to the admin.

Table 2.15: Manage Users Use Case

2.1.16 Monitor System

Field	Details
Use Case Name	Monitor System
Scope	The admin monitors the system to ensure smooth operation, track performance, and detect any issues.
Level	Admin Goal
Primary Actor	Admin
Stakeholders and interests	Admin: Wants to ensure the system operates efficiently and detects any performance or security issues. System: Needs to provide real-time data and logs for monitoring system health and activity.
Preconditions	The admin is authenticated and has the necessary permissions The system is operational and provides access to logs, performance data, and alerts.
Post-conditions	The admin successfully monitors system performance and is alerted to any issues (if applicable).
Main Success Scenario	1. The admin logs into the system and navigates to the monitoring dashboard. 2. The system displays performance measures, such as server health and response times. 3. The admin monitors logs, metrics, and alerts for any anomalies or system errors. 4. The admin takes action based on the logs.

Table 2.16: Monitor System Use Case

2.2 Functional Requirements

This section describes the functional requirements of VocalCraft.

2.2.1 User Authentication

The following are the requirements for User Authentication:

1. The users will sign up using their email, password, and username.
2. The system will authenticate the user.
3. The system shall allow users to log out of their accounts.

2.2.2 Voice-to-Text Processing

The following are the requirements for Voice-to-Text Processing:

1. The system shall use the user's microphone for voice input.
2. The system shall convert the voice input into text using a speech recognition model (Whisper).
3. The system shall display the text to the user for confirmation or editing.

2.2.3 Text Input

The following are the requirements for Text Input:

1. The system shall allow users to input text prompts for poster generation manually.
2. The system shall validate the text input to ensure it is not empty and make sure no information is missing (e.g., background color).

2.2.4 Poster Generation

The following are the requirements for Poster Generation:

1. The system shall generate a poster based on the user's voice or text prompts using an image generation model (Stable Diffusion).
2. The system shall present the generated poster to the user for preview.

2.2.5 Poster Editing

The following are the requirements for Poster Editing:

1. The system shall allow users to edit the generated poster (e.g., move text, change colours, and fonts, and resize elements).
2. The system shall apply user edits to the poster in real-time.
3. The system shall allow users to undo changes made during the editing process.

2.2.6 Save and Load Prompts

The following are the requirements for Save and Load Prompts:

1. The system shall enable the users to save their voice or text prompts for later use.
2. The system shall allow users to retrieve previously saved prompts and reuse them to generate posters.

2.2.7 Save and Share Posters

The following are the requirements for Save and Share Posters:

1. The system shall permit the users to download generated or edited posters in conventional formats such as PNG or JPEG.
2. The system shall allow users to share the generated posters directly via social media, email, or messaging apps.

2.2.8 System Monitoring and Management (Admin Features)

The following are the requirements for System Monitoring and Management:

1. The system shall enable an admin to monitor system performance of such functions as usage of the poster generation and user activity.
2. The system shall enable an admin to create user profiles, delete and modify users' access permission.
3. This system shall be capable to log errors and enable the admin to view log in case of troubleshooting.

2.3 Non-Functional Requirements

This section specifies the nonfunctional requirements of VocalCraft.

2.3.1 Reliability

1. It is required that the Mean Time Between Failure (MTBF) should not be below 1000 hours of uninterrupted usage.

2. They defined a system failure as the inability to produce or generate a poster, respond to user input, or operate with voice/text prompts.
3. If failure happens, the system should be able to recognize that or at least recognize the right time to alert the admin and the user.
4. The system must have automatic recovery mechanisms such as retries for poster generation or a fallback strategy in case of a system failure.
5. The system has to produce and store failure logs.
6. The admin needs to be informed of consecutive failures that occur and records need to be kept of all system failures to analyze further.

2.3.2 Usability

1. **USE-1:** The system shall allow users to generate a poster from a voice or text prompt with no more than 3 interactions.
2. **USE-2:** The system's interface must allow users to preview the generated poster and make edits (e.g., adjust text or colors) with drag-and-drop functionality.
3. **USE-3:** The system must provide real-time visual feedback during poster generation, including progress bars or loading indicators.
4. **USE-4:** The system shall allow users to undo or reset any poster edits with a single interaction.
5. **USE-6:** The system must provide tips for first-time users to help them understand the main features.

2.3.3 Performance

1. **PER-1:** 95% of posters generated by the system shall be completed within 30 seconds from the time the user submits the request, under normal operating conditions.
2. **PER-2:** 90% of voice-to-text conversions shall be processed and displayed to the user within 5 seconds after submission of voice input.
3. **PER-3:** The system must support at least 100 concurrent users generating posters without noticeable performance degradation (response times not exceeding 30 seconds).

4. **PER-4:** The system shall allow the user to save or share posters within 5 seconds after the user confirms the poster is finalized.
5. **PER-5:** In case of heavy load (more than 100 concurrent users), the system must maintain a response time of less than 60 seconds for 80% of poster generation requests.

2.3.4 Security

1. **SEC-1:** The system shall prevent unauthorized access to user data and poster generation history, requiring a high level of technical skill and significant time to breach.
2. **SEC-2:** The system shall detect and log any unauthorized access attempts and notify the admin within 5 minutes of the detection.
3. **SEC-3:** The system shall ensure that user data is protected against unauthorized modifications, and any changes to sensitive data must be traceable.
4. **SEC-4:** The system shall maintain the integrity of poster data during transmission and storage to prevent data corruption.

2.4 Domain Model

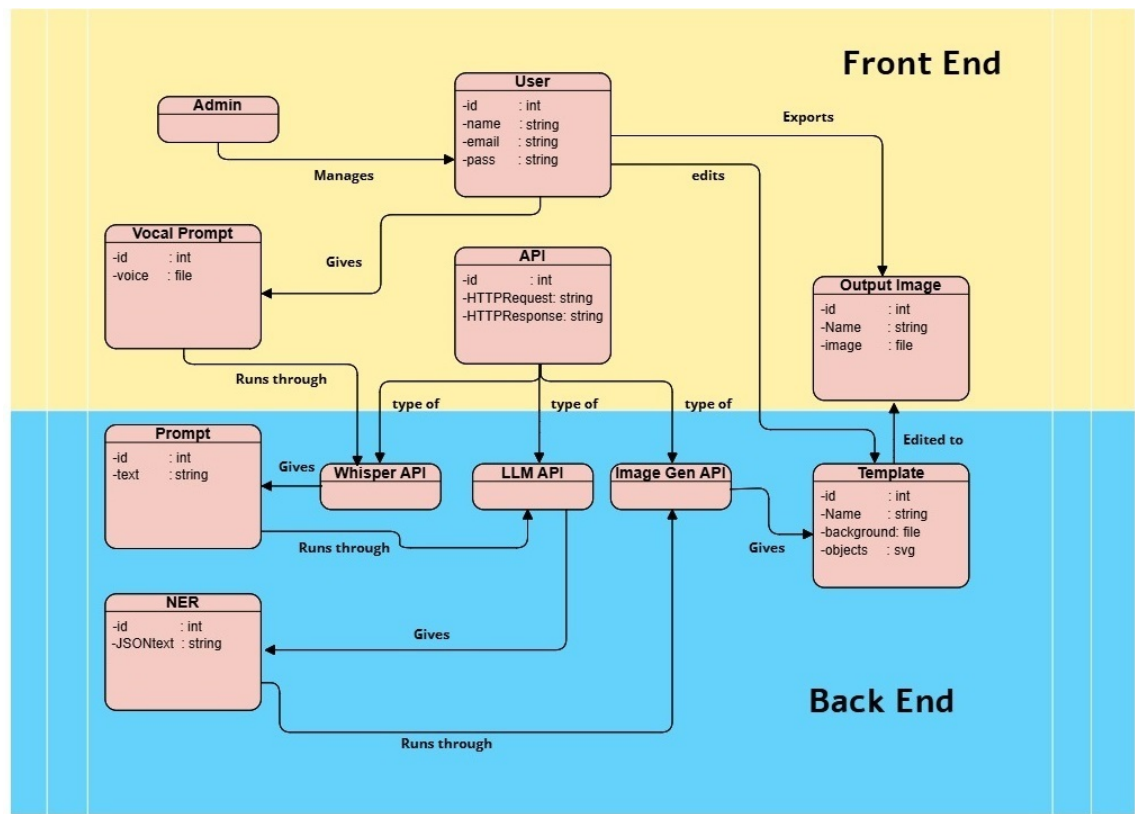


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

The architectural design followed by VocalCraft will be API-Centric Microservices with Event-Driven Workflow. This architecture offers flexibility, maintainability, and scalability. Most of the core functionalities of the app are abstracted within the external APIs and the app provides a procedural event-based workflow to the user. Event-based ensures smooth asynchronous communication between services, allowing the user real-time interaction with the app. This architecture is also secure due to the API gateways, and token-based authentication that protects the sensitive operations of the app and user's data.

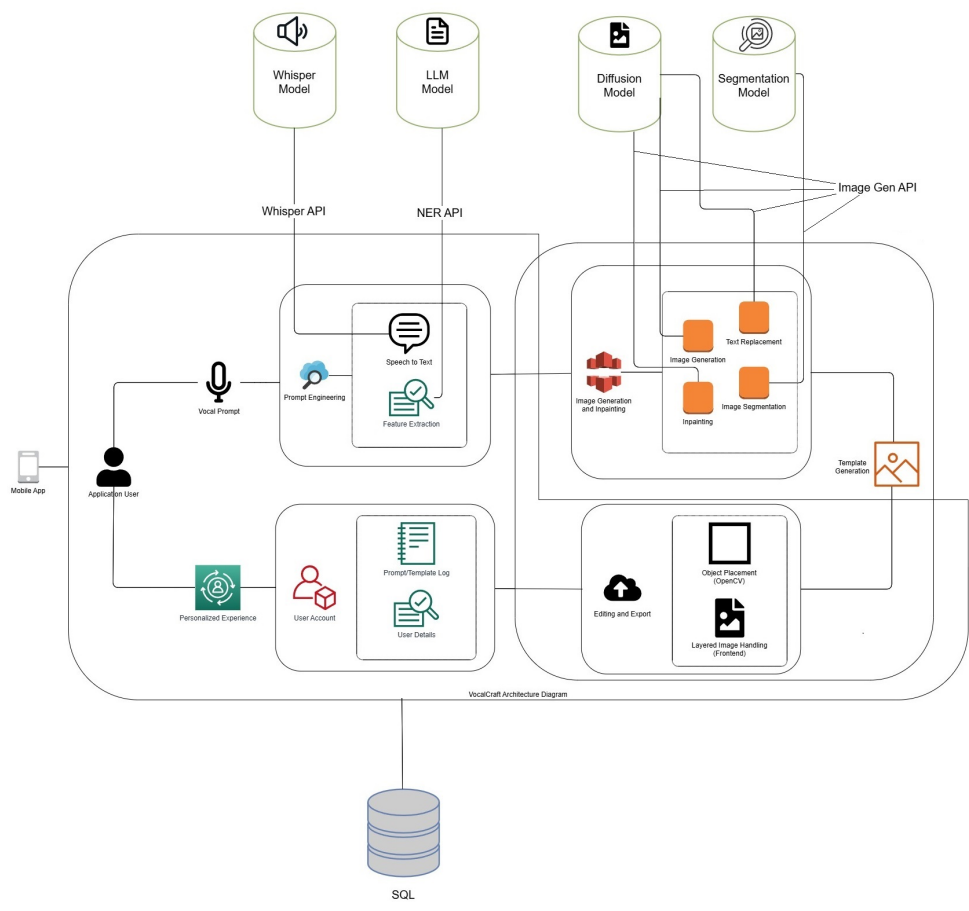


Figure 3.1: Architectural Design Diagram

3.2 Design Models

3.2.1 ERD Diagram

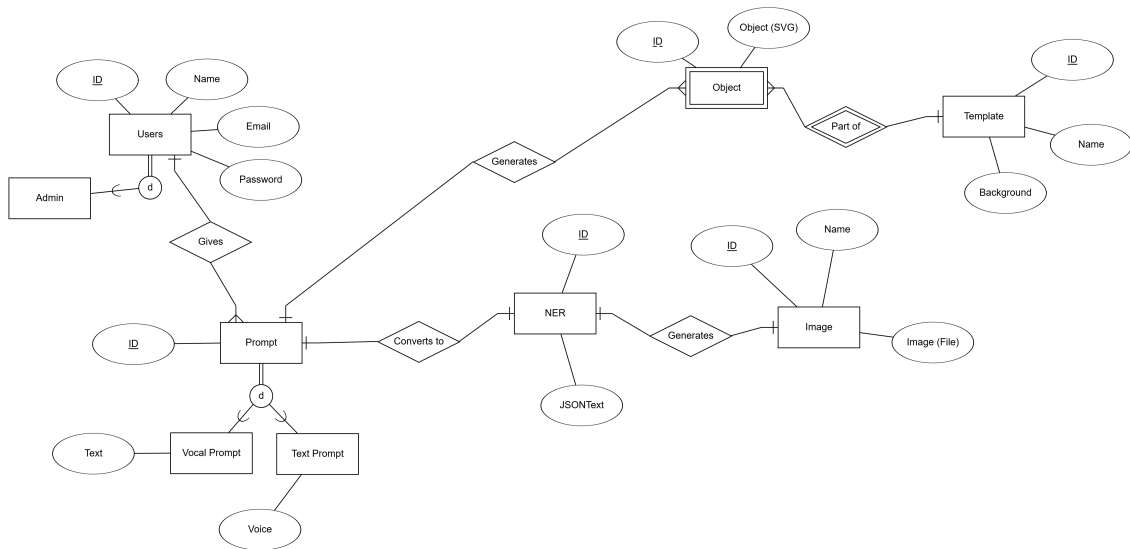


Figure 3.2: ERD Diagram

3.2.2 Activity Diagram

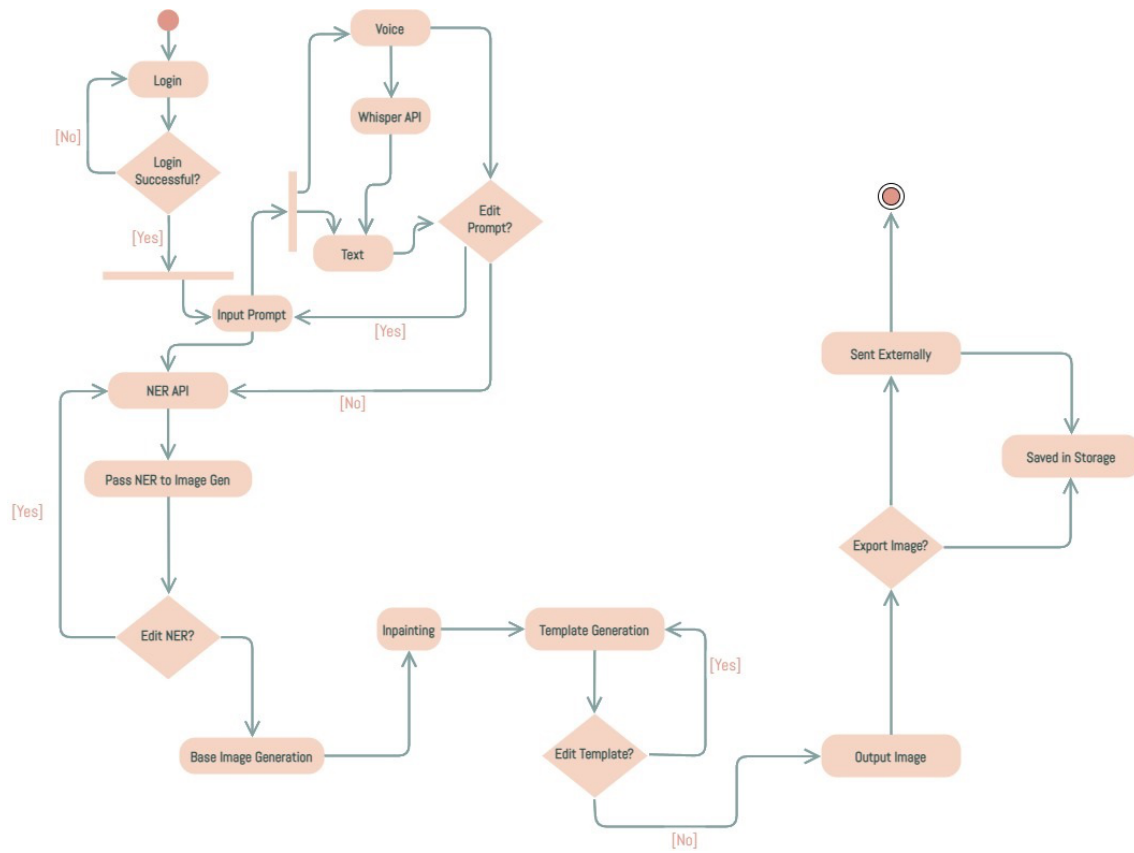


Figure 3.3: Activity Diagram

3.2.3 Data Flow

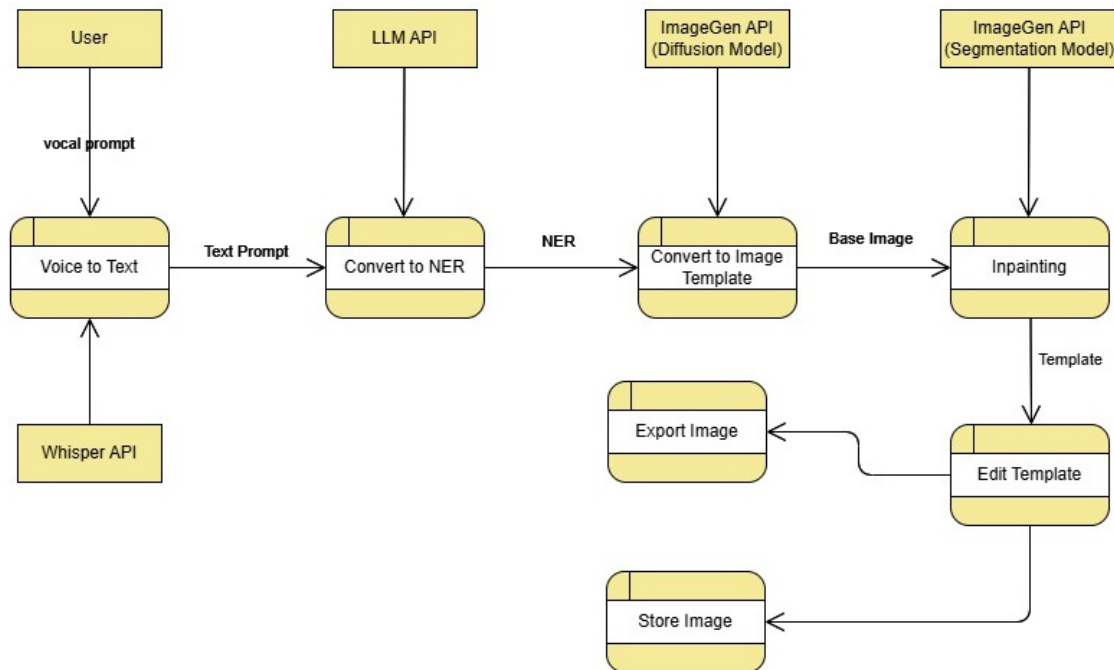


Figure 3.4: Data Flow Diagram

3.2.4 State Transition Diagram

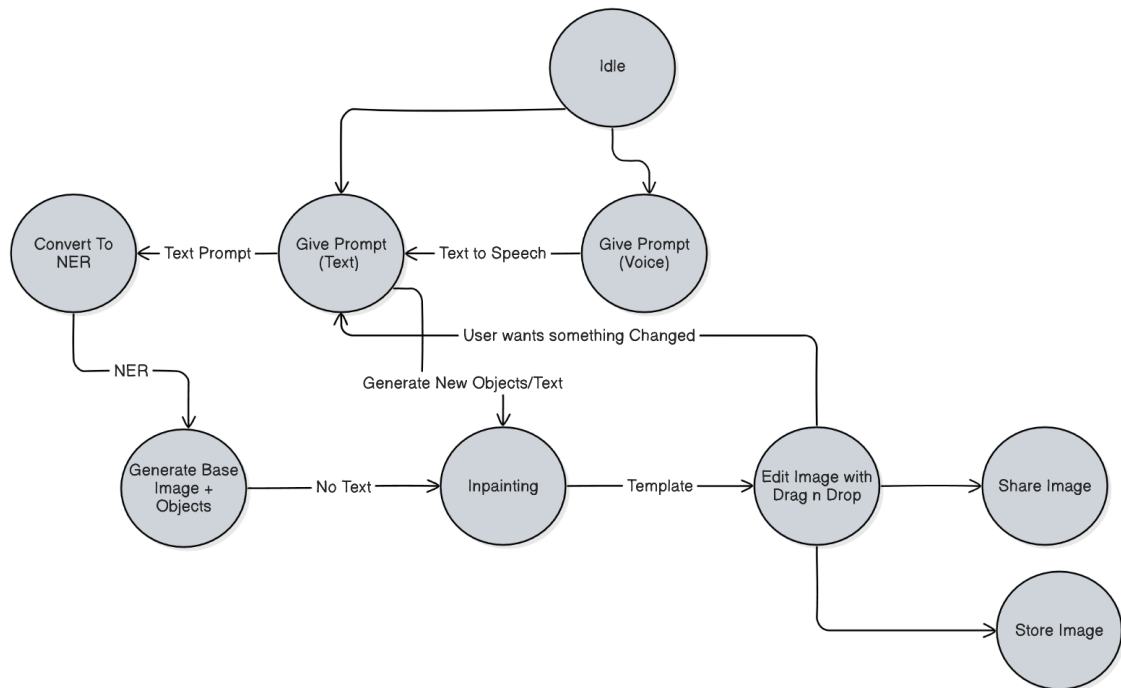


Figure 3.5: State Transition Diagram

3.2.5 System Sequence Diagrams

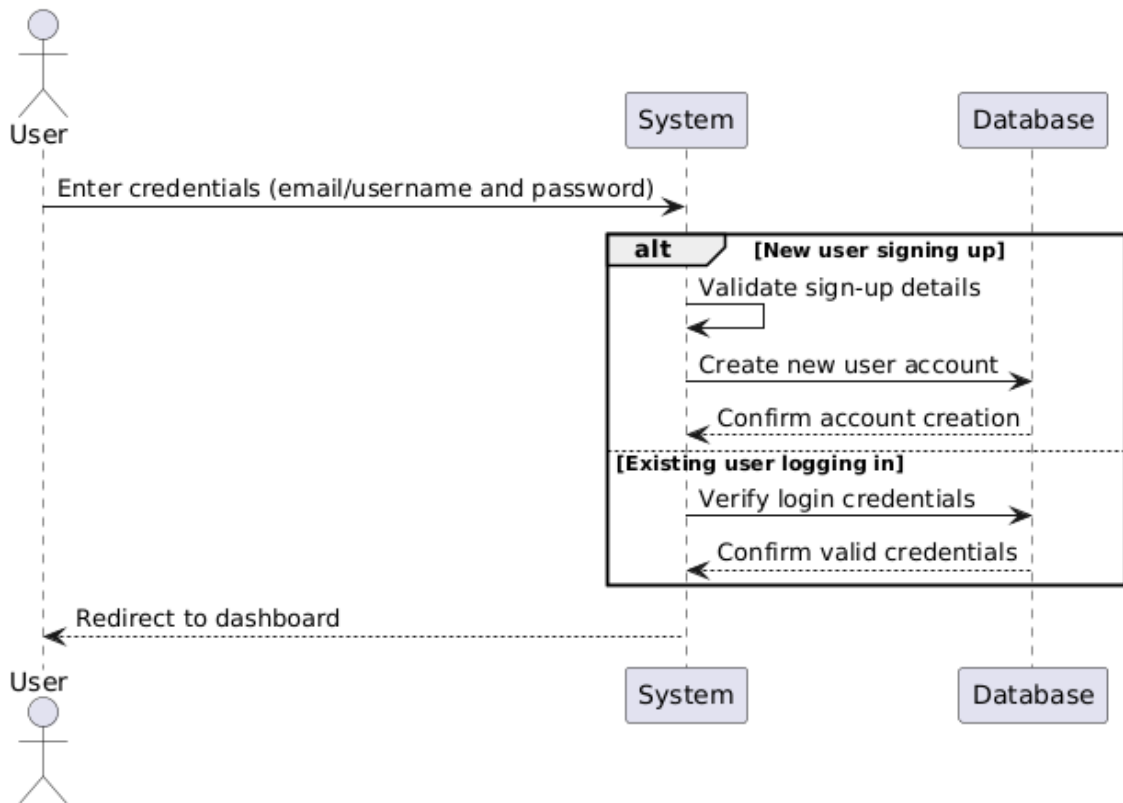


Figure 3.6: System Sequence Diagram for User Authentication Sign Up/Log In

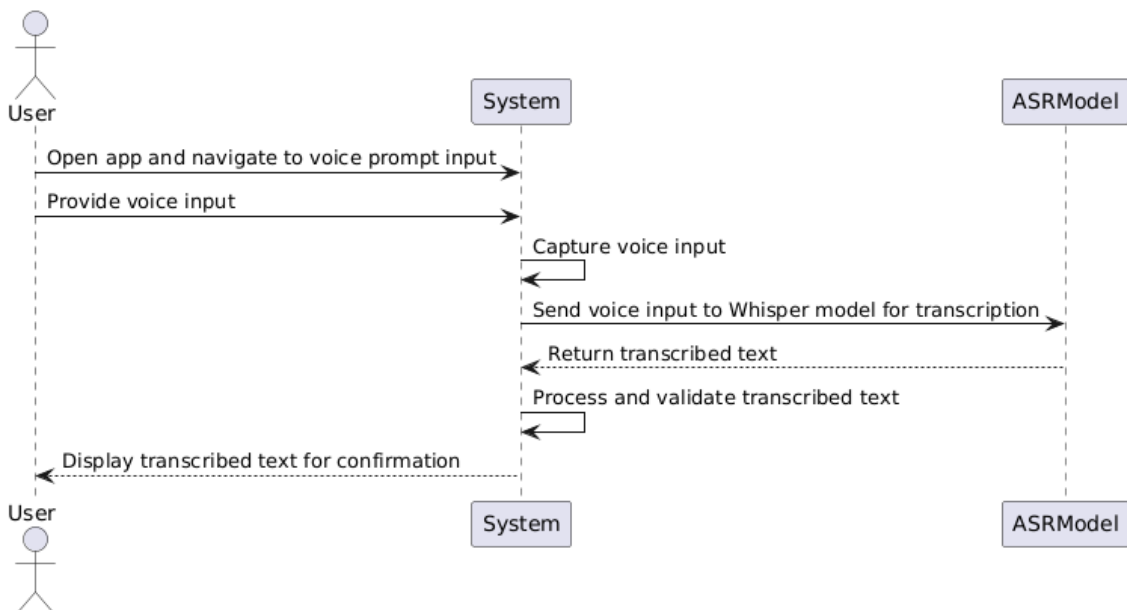


Figure 3.7: System Sequence Diagram for Providing Voice Prompt

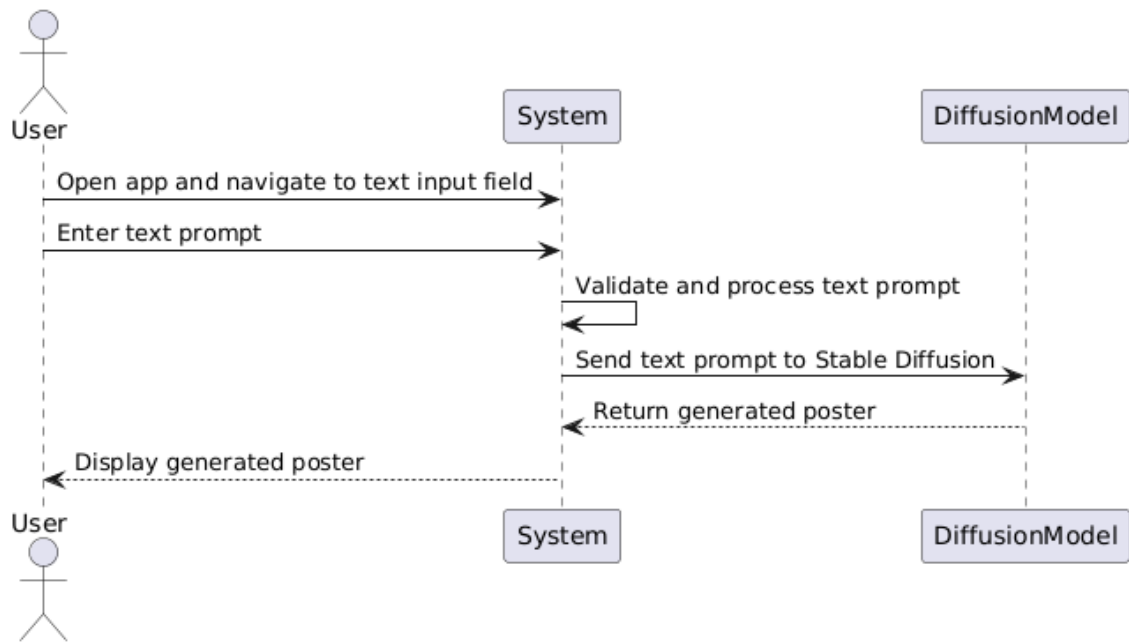


Figure 3.8: System Sequence Diagram for Providing Text Prompt

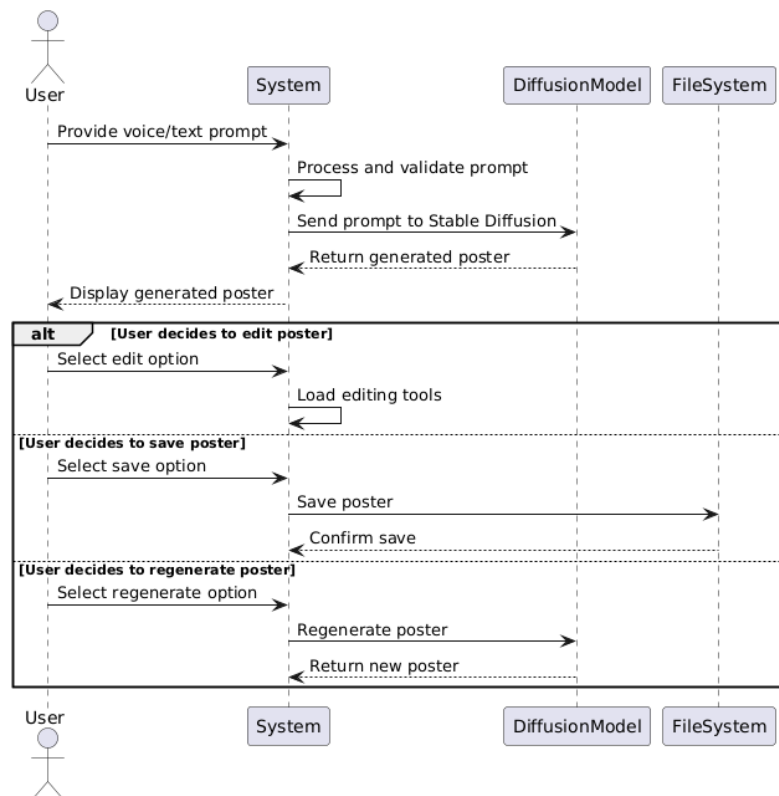


Figure 3.9: System Sequence Diagram for Generating Poster

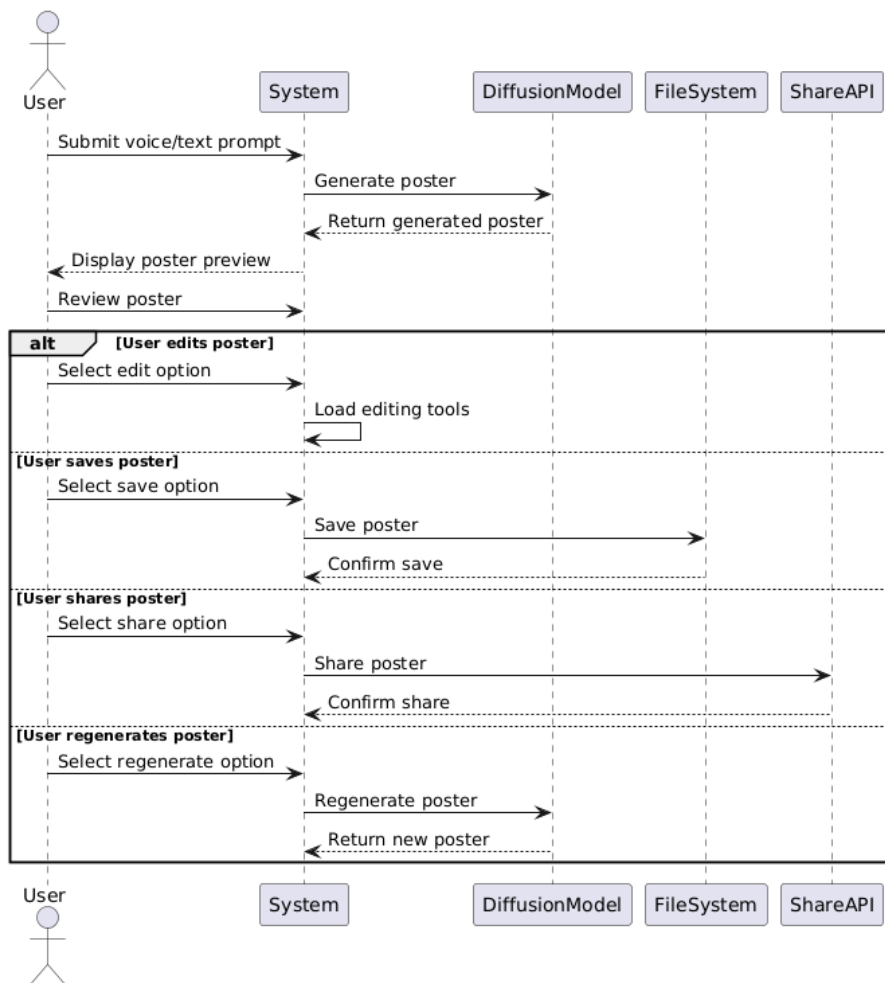


Figure 3.10: System Sequence Diagram for Preview Poster

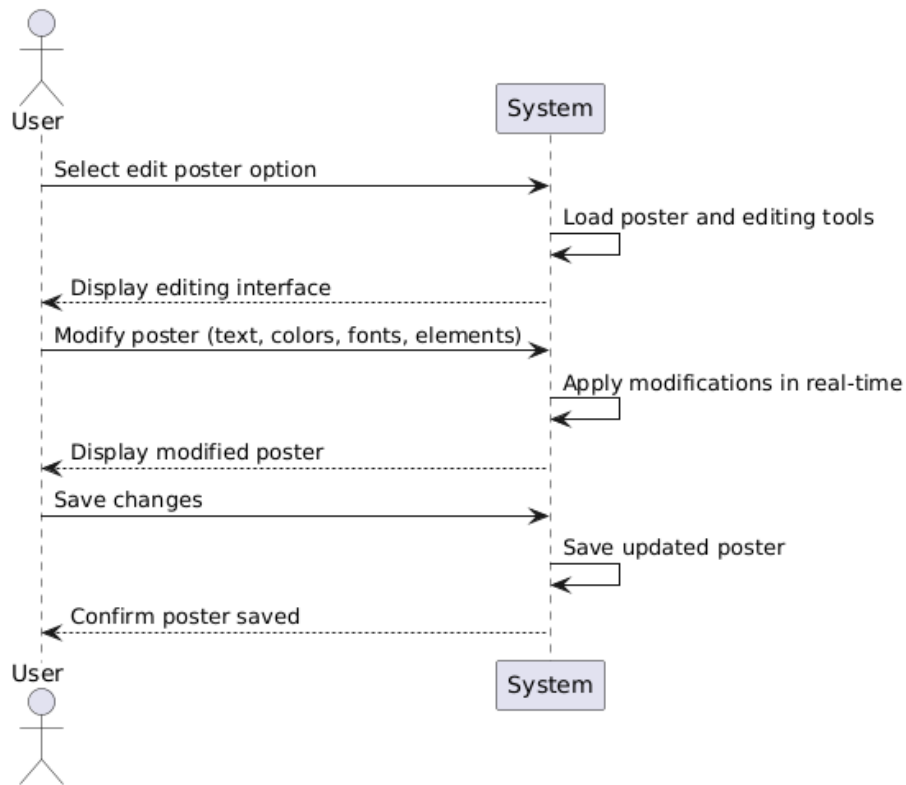


Figure 3.11: System Sequence Diagram for Edit Poster

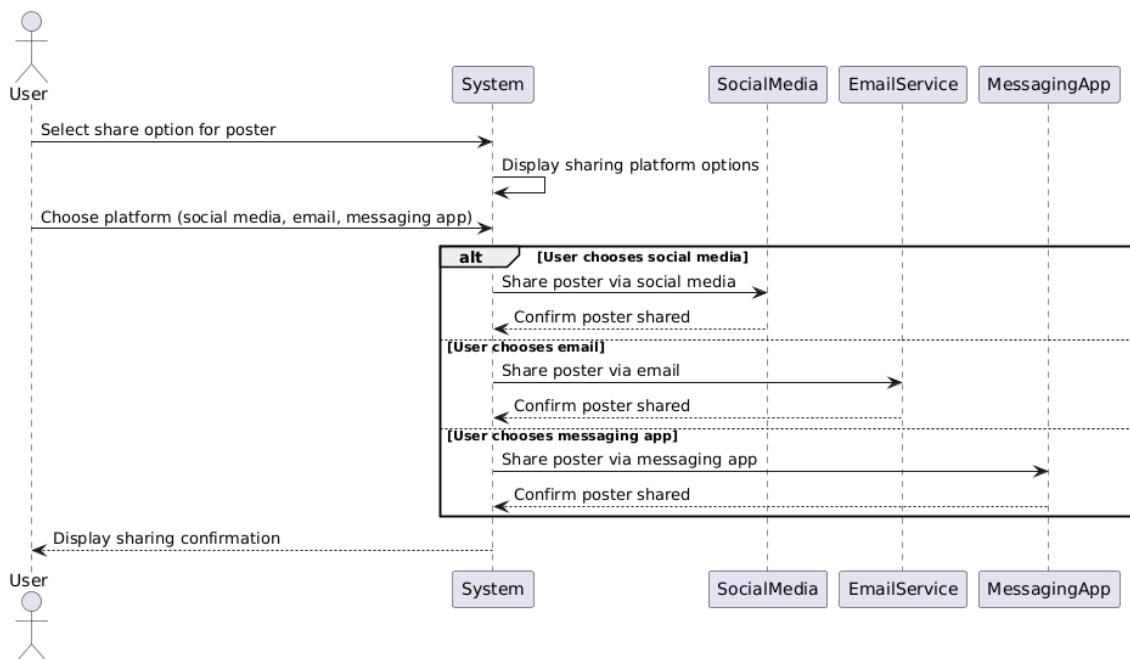


Figure 3.12: System Sequence Diagram for Share Poster

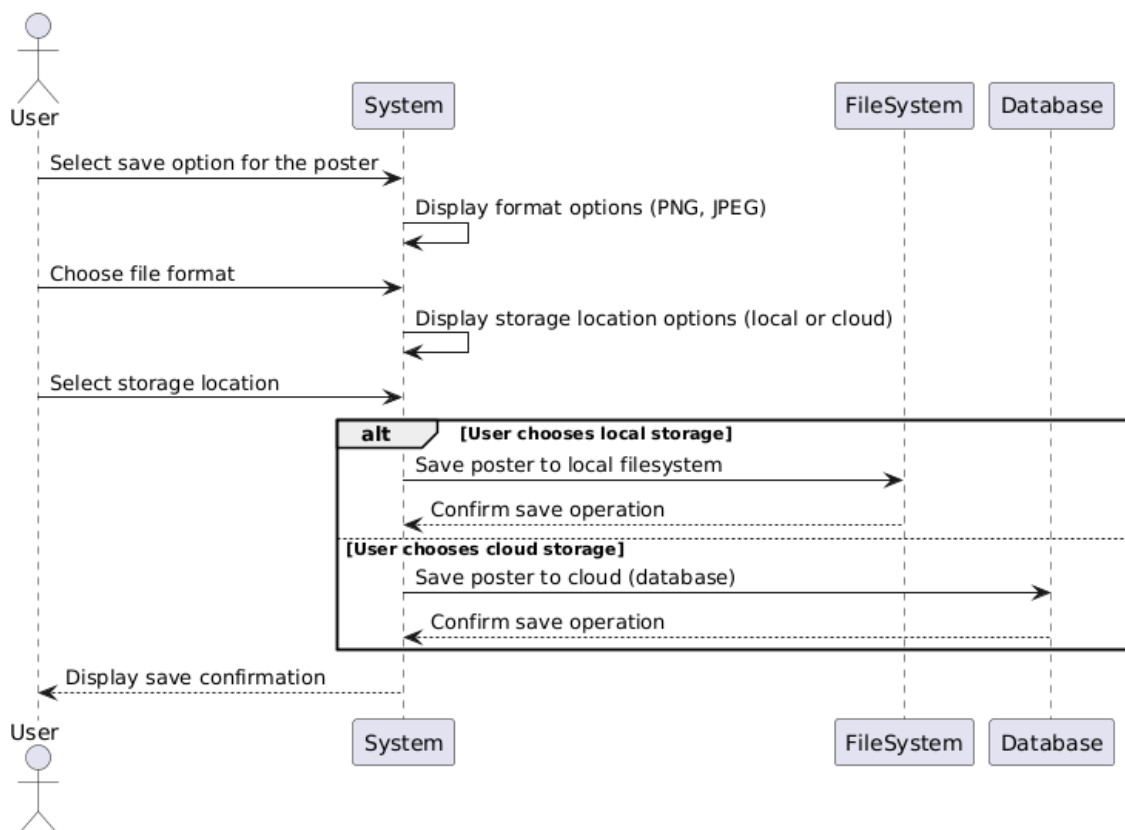


Figure 3.13: System Sequence Diagram for Save Poster

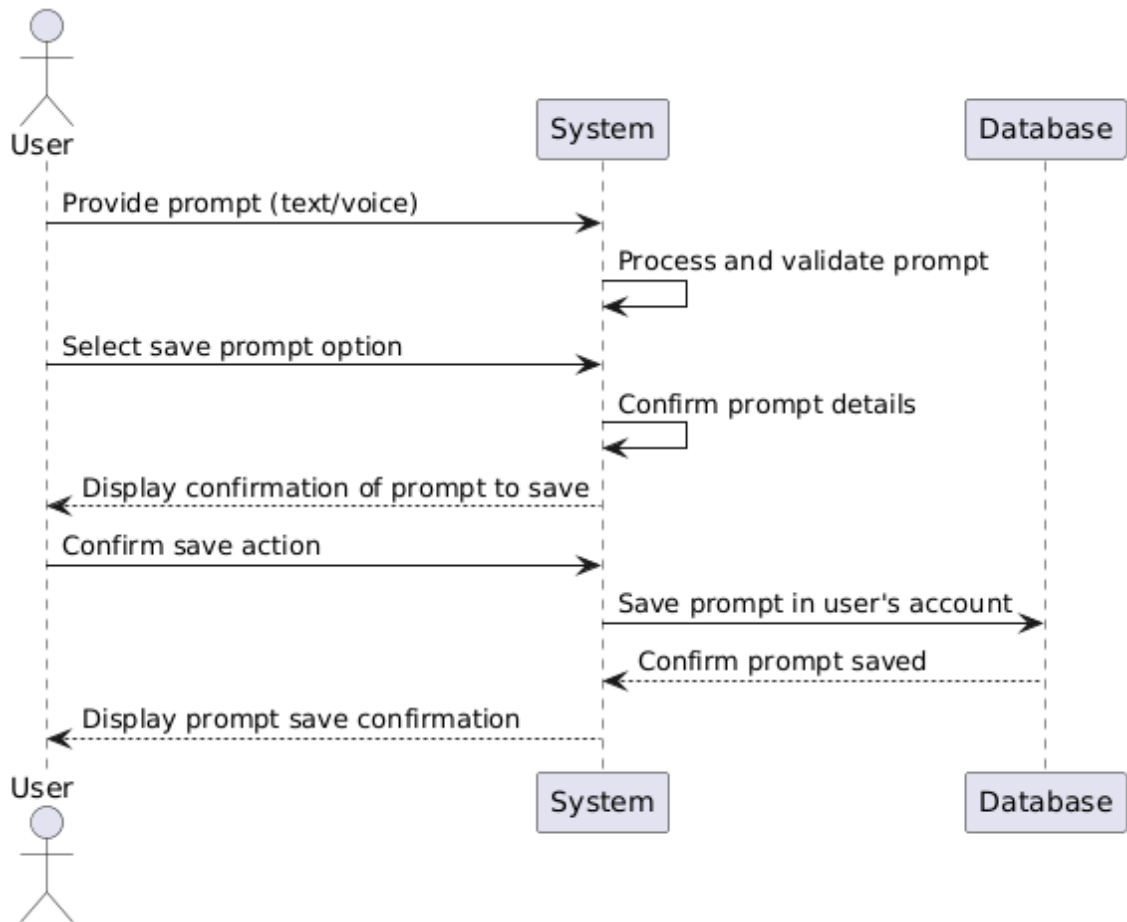


Figure 3.14: System Sequence Diagram for Save Prompts

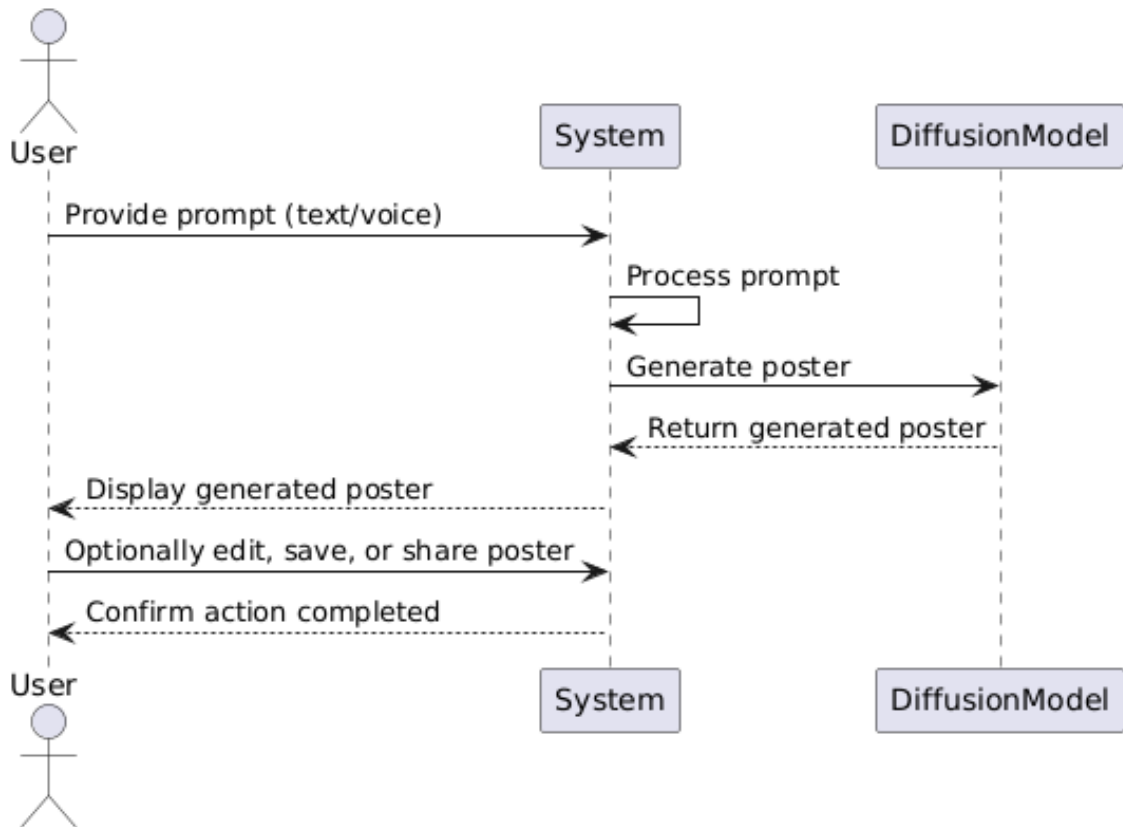


Figure 3.15: System Sequence Diagram for Return poster to user

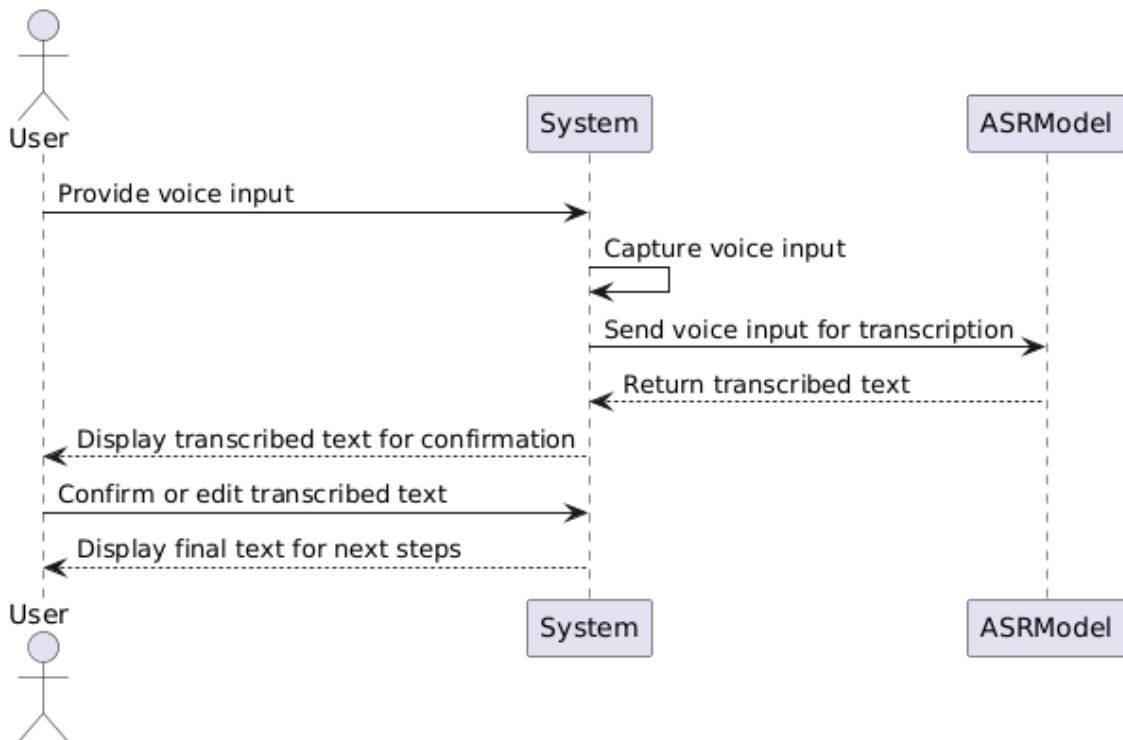


Figure 3.16: System Sequence Diagram for Process Voice Input

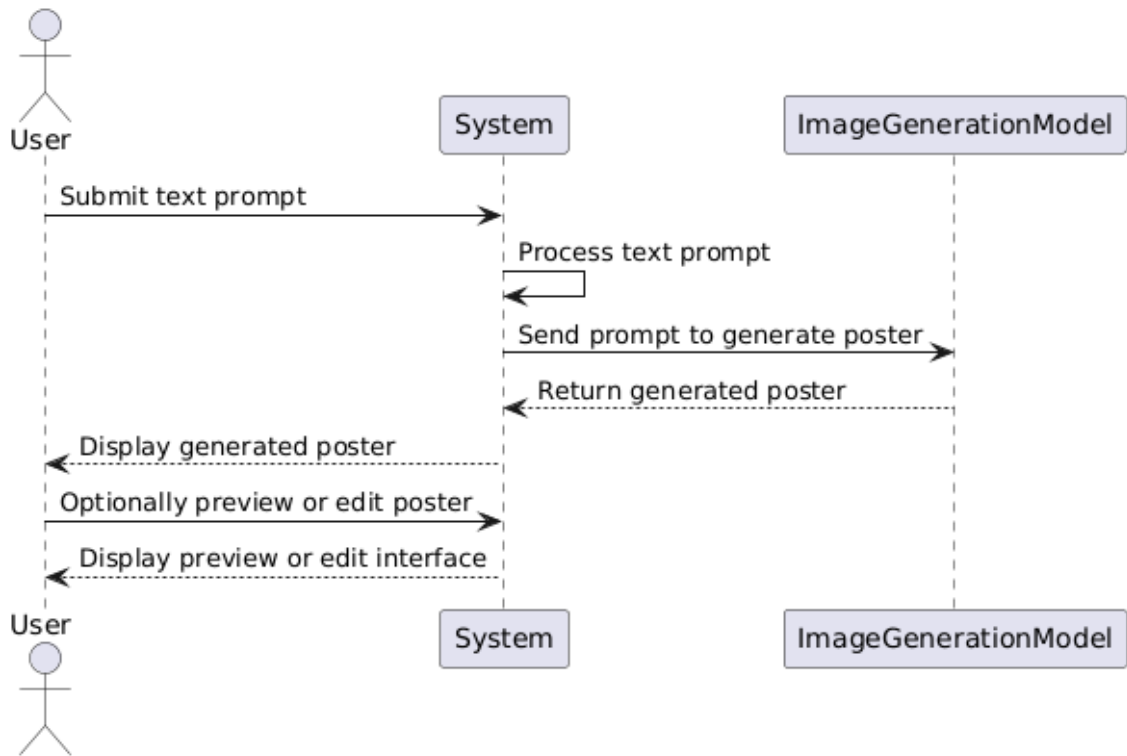


Figure 3.17: System Sequence Diagram for Generate Poster From Text

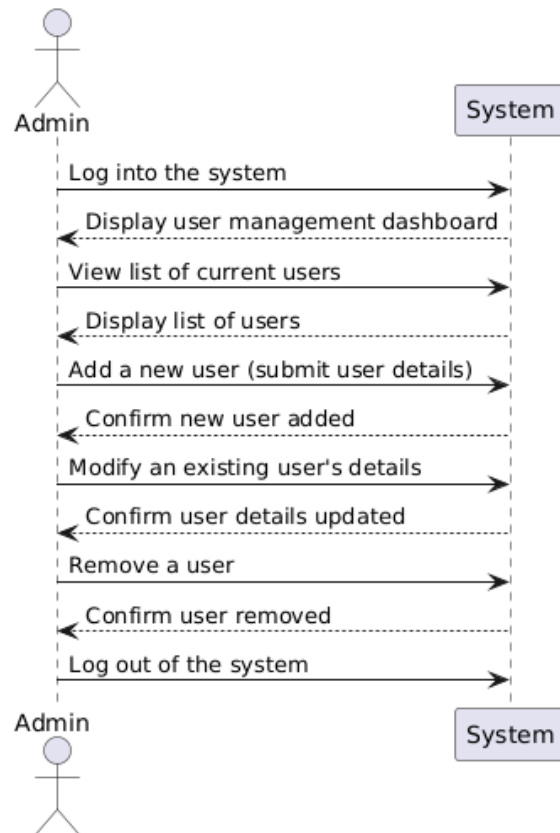


Figure 3.18: System Sequence Diagram for Manage Users

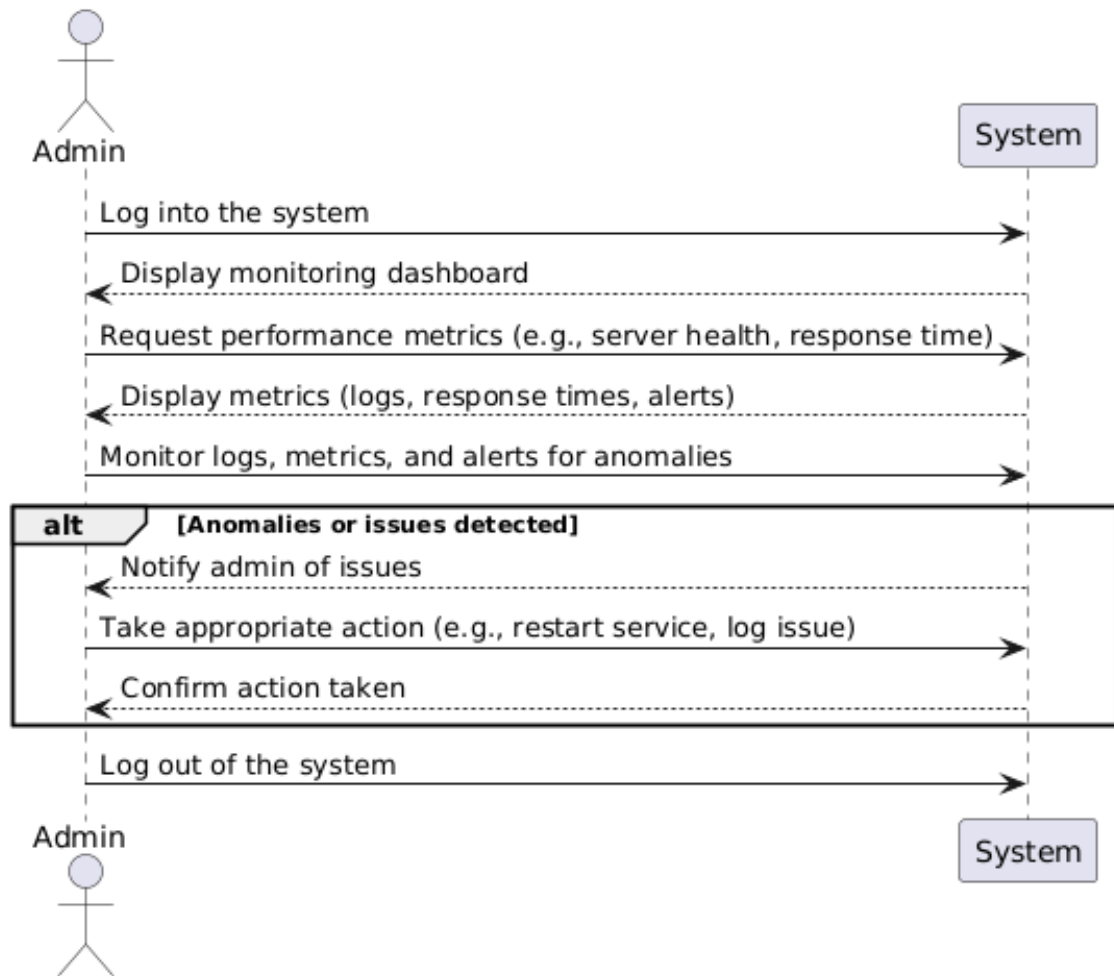


Figure 3.19: System Sequence Diagram for Monitor System

3.3 Data Design

3.3.1 Information Domain Transformation

It verifies the user's credentials for authentication. Then the user speaks to the system or writes a text input, and the system translates it into the pamphlet of the user requirement. The input is then passed through external API such as STT conversion, prompt engineering, JSON prompts file, image generation, and image inpainting. The data generated is used to construct the image, and the user can also share and or download the image.

1. System Inputs

- User Login Credentials (Text Input)
- Image Prompt (Voice/Text Input)

2. System Output

- Generated Image
- Sharing and Downloading options

The app converts the voice prompt to text using an external speaker-to-text translation application. Once in text form, the given textual prompt is refined, and transformed into a JSON file using the Prompt Engineering and the NER API which defines key entities. Later, the prompt is transferred to the Diffusion API to generate the image and its parts in order to obtain the result.

3.3.2 Data Structures

The data entities in the app are for the user entries, user prompts, and generated images.

1. User Entity

- **Attributes:** ID, Name, Email, Password
- **Description:** Stores the user login and signup information

2. User Prompts

- **Attributes:** ID, Prompt
- **Description:** Stores the user prompts.

3. Image Entity

- **Attributes:** ID, Name, ImageData, DateCreated, ShareData
- **Description:** Stores metadata for each image generated by the user, including whether it was shared or downloaded.

Relationships:

- There is a **one to many** mapping between entities **User** to **Image** because one instance of a user can provide many images.
- A **one-to-many** relationship exists between **User** and **User Prompts** since 1 user can generate many prompts.
- A **one-to-one** relationship exists between **User Prompts** and **Image** entities, as one prompt by a single user will generate only one image.

3.3.3 Storage Mechanisms

The app uses a relational database to store user credentials and image data securely.

- **Database:**
 - **Type:** Relational Database (MySQL)
 - **Purpose:** To store user login details, user prompts, and generated images.
- **Storage Strategy:**
 - **User data** is stored in a secure table with hashed passwords.
 - **Prompt Data** is stored in the table linked with the User table, organized by time created, and could be filtered out for each user.
 - **Image data** is stored along with metadata, including creation date and user ID for easy retrieval.

3.3.4 Data Processing

After the user login, the app processes the input prompt through several stages:

- **Voice-to-Text Conversion:** The voice input is then passed to an API outside this system where it is converted into text.
- **Prompt Engineering and NER Processing:** The subsequent text is processed using a Prompt Engineering and a Named Entity Recognition API to perform Key Entity Extraction.

- **Image Generation:** The extracted entities are then employed to create an image via another API.

Concurrency and Transactions:

- Transactions for user login and image generation are handled in a sequential process, ensuring each stage is completed before moving to the next.
- **Session management** is used to maintain user states, ensuring they remain logged in during the process.

3.3.5 Security and Integrity of Data

Data security is maintained throughout the app, particularly for sensitive user information and images.

- **Data Protection:** User passwords are securely hashed, and sensitive data is encrypted in storage.
- **Access Control:** Before accessing the app's features, user authentication will be required. Users and an admin will have different accesses in the app.

3.3.6 Data Flow Between Subsystems

The app consists of several subsystems:

- **User Authentication Subsystem:** Handles user login and sign up.
- **STT Subsystem:** Transforms voice prompts into text using an STT API.
- **Prompt Engineering and NER Subsystem:** Extracts entities set in a pre-defined prompt layout from the text prompt.
- **Image Generation Subsystem:** Creates images and components within the images based on extracted entities using an external API.
- **Image Sharing/Download Subsystem:** Allows users to share or download the generated image.

These subsystems follow a sequential data flow from getting the user input prompt to generating the pamphlet.

3.3.7 Optimization and Performance Considerations

The app's performance will be a result of the following optimizations:

- **Asynchronous API Calls:** API calls for voice-to-text conversion and NER processing are handled asynchronously to reduce latency.

Chapter 4

Conclusion

In conclusion, through the application of voice-to-text processing, prompt engineering, and diffusion models, the **VocalCraft** will showcase its capability to completely transform the way promotional pamphlet is done. Being an app for mobile use and having customizable templates and cloud storage, VocalCraft offers fast-food businesses an effective and easily affordable tool for their marketing work.

Such voice input and text output modules, along with the image processing module guarantee the attractiveness and, at the same time, professional appearance of the text presented in the proposed pamphlets. Concerning the current market, the project directly responds to some of the principal problems encountered with graphic design tools because it provides straightforward software that does not require any prior experience in design.

Also as VocalCraft gradually transitions to a real-world application, there will be a need to improve the performance, accessibility, and security of the system as it forms a competitive tool for the marketing industry.

Glossary

ASR (Automatic Speech Recognition): A technology that converts spoken language into text using algorithms and machine learning models. In this report, it refers to the use of the Whisper model to transcribe voice input.

Diffusion Model: A generative model that generates images by iterative denoising a random noise image based on a given input prompt. Used for creating images in VocalCraft's poster generation process.

NER (Named Entity Recognition): A process in natural language processing (NLP) that identifies and classifies key information (entities) such as names, dates, and locations from text data. In this project, it helps extract relevant details from user input to generate the pamphlet content.

Prompt Engineering: The process of designing input prompts for AI systems to guide their output. In this project, the Prompt Engineering will be done on the textual prompt for the diffusion models to create a pamphlet.

Inpainting: A technique used to fill in missing or masked parts of an image. In the VocalCraft project, inpainting is used to add placeholders for text and object areas within the generated image.

STT (Speech-to-Text): A technology that converts spoken language into written text. Whisper, an STT model, will be used in this project to transcribe the user's voice input into text.

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